KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ART AND BUILT ENVIRONMENT

(BSc.) Architecture, Land Economy, Quantity Surveying and Construction Economics, Real Estate and Settlement Planning

FIRST SEMESTER EXAMINATIONS, 2022

GE 183 PRINCIPLES OF LAND SURVEYING

First Year

SET A

May 2022 2 hours

Index Number: .....Signature:.....

Programme of Study: .....Year: ...

Instructions

i. Answer **All Questions** on the multiple-choice answer sheet by shading the correct answer from the alternatives lettered **'a to e'** and also circle the answer on the Question Paper.

ii. Submit the question paper and the multiple-choice answer sheet (**Scannable Sheet**) after the exam.

1. Surveying is an art and science of ..... and the representation of these information either graphically or numerically?

   i. Making measurements of relative positions of natural features only
   ii. Making measurements of relative positions of natural and artificial features
   iii. Making measurements of relative positions of artificial features only

   a. i, ii & iii
   b. i & ii only
   c. ii & iii only
   d. ii only
   e. None of the above

2. The purpose of Surveying is/are to?

   i. Setting out on the ground of the positions of proposed construction or engineering works
   ii. Measure features on the land so that they can be accurately plotted on a map
   iii. Setting out angles and distances of boundary lines

   a. i & ii only
   b. i & iii only
   c. ii & iii only
   d. i, ii & iii
   e. None of the above

3. The preliminary Inspection of the area to be surveyed is known as?

   a. Rough survey
   b. Primary survey
   c. Route survey
   d. Reconnaissance survey
   e. None of the above

4. The cardinal principle in Land Surveying is to work from?
   a. Higher level to a lower level
   b. Lower level to a higher level
   c. Part to whole
   d. Whole to part
   e. None of the above

5. The main aim for enforcing the practice in **Question 4** is to?

   a. To estimate the total area of the land correctly
   b. Make fieldwork easy
   c. Localize the errors and prevent their accumulation
   d. Enable the Surveyor to fix boundary points
   e. None of the above

a. It is easy to read
b. It is not tied to a particular unit
c. It remains correct if the map is reduced or enlarged photographically
d. It is more accurate than the other types
e. None of the above

26. Which of the following is/are true about the plan scale?

i. It is the ratio between a distance on a map and the corresponding distance on the terrain.
ii. It is also an expression of how much the area represented has been reduced on the map
iii. It shows the relationship between two distances either on a map or on the ground

a. i & ii only
b. ii & iii only
c. i, ii & iii
d. i & iii only
e. None of the above

27. What is the ground area of a plot of land which has its plan area as 0.80 mm^2 when it is plotted on a plan of 1: 5000?

a. 20000 m^2
b. 200 m^2
c. 20 m^2
d. 2000m^2
e. None of the above

28. A plot of land is in the form of a triangle in which the base is two times the height. When surveyed, it was found to have an area of 15,000 m^2. Calculate the base of the land as drawn on the plan at a scale of 1:500?

a. 360.00 mm
b. 244.49 mm
c. 346.00 mm
d. 489.90 mm
e. None of the above

29. A lake was measured on a plan and found to cover an area of approximately 8cm by 12.5cm. Given that the scale of the map is 1:2500, calculate the ground areas in hectares? **NB:** 1 cm^2 = 1e-8

a. 0.6250
b. 6.250
c. 62.50
d. 625
e. None of the above

b. measured value to its mean value.
c. measured value to its weighted mean value.
d. repeated measurements of the same quantity to each other.
e. None of the above

12. Accuracy is a term that indicates the degree of conformity of a measurement to its:
    a. most probable value
    b. mean value
    c. true value
    d. standard error
    e. None of the above

13. Systematic errors.....

    i. Have the same sign and magnitude
    ii. Remain even after all error types have been removed
    iii. Occur easily

    a. i, ii, iii are all true
    b. i and ii are true
    c. Only i is true
    d. i and iii are true
    e. i, ii, and iii are all false

14. Random Errors:

    i. Remain in measurements after only mistakes have been eliminated
    ii. The magnitude and sign are not constant
    iii. They tend to partially cancel themselves in a series of observations

    a. i and ii only
    b. i and iii only
    c. ii and iii only
    d. i, ii, and iii
    e. None of the above

15. One of the characteristics of random errors is that ?
    a. Small errors occur as frequently as the large errors
    b. Positive errors occur more frequently than the negative errors
    c. Small errors occur more frequently than the large errors
    d. Large errors may occur more frequently
    e. None of the above

16. Transposition error is an example of .....

    a. Random error
    b. Gross error
    c. Systematic error
    d. Compensating error

b. N 90° 00′ E
c. S 05° 25′ E
d. S 84° 35′ W
e. None of the above

69. Calculate the angle between AB and AC?
a. 095° 25′
b. 259° 35′
c. 164° 35′
d. 084° 35′
e. None of the above

70. The clockwise angle measured from the true north line to a survey line is known as?
a. Magnetic bearing
b. Arbitrary bearing
c. True bearing
d. Compass bearing
e. Grid bearing

71. What is the whole circle bearing of N 20° 45′ E?
a. 20° 45′
b. 160° 45′
c. 339° 15′
d. 340° 45′
e. 159° 15′

72. Given that the magnetic declination at a place is 6° 30′ W, what is the true bearing of a line AB whose magnetic bearing is 155° 50′?
a. 149° 20′
b. 162° 20′
c. 296° 20′
d. 315° 20′
e. 99° 50′

Use the information to answer questions to 73 to 100.

A client has asked for an as-constructed drawing of a recently built industrial building. As part of the survey the following series of levels were booked and reduced on the field. The surveyor carelessly dropped the book into a ditch such that some of the entries were obliterated. With the provided information, refill the table.

e. None of the above

Use the information below to answer questions 17 and 18

i. Systematic errors
ii. Random errors
iii. Mistakes

17. Which of the above error(s) follow the normal distribution curve?

a. i only
b. ii only
c. iii only
d. i and ii only
e. ii and iii only

18. Which error(s) would not be eliminated by repeated measurements under the same conditions and using the same equipment?

a. i only
b. ii only
c. iii only
d. i and ii only
e. none of the above

19. Which of the following is/are correct about "True" Value?

i. The "true" value is determined statistically after repeated measurements.
ii. The "true" value for a distance is taken as the mean value for a series of repeated measurements
iii. The "true" value can be obtained by taking only one accurate measurement

a. i only
b. ii only
c. i & ii
d. i, ii & iii
e. None of the above

Use the information below to answer questions 20 and 21

i. True value
ii. Precision
iii. Accuracy

20. Since the ..... is seldom known, ... is generally indeterminate in practice.

a. i & ii respectively

b. i & iii respectively
c. ii & iii respectively
d. ii & i respectively
e. None of the above

21. ..... refers to the degree of refinement or consistency of a group of observations

a. i
b. ii
c. iii
d. i & ii
e. None of the above

22. A Graphic /Bar scale:

i. is Probably the most common type of scale found on maps
ii. is not correct if the map is reduced or enlarged photographically
iii. is usually short compared to the map itself

a. i and ii only
b. i and iii only
c. ii and iii only
d. i, ii, and iii
e. None of the above

Use the information given to answer Questions 23 and 24.

i. Representative Fraction
ii. Verbal Scale
iii. Graphical scale

23. Which of the above is/are likely to introduce errors as a result of enlargement or reduction of the map?

a. i only
b. ii only
c. iii only
d. i, ii and iii only
e. None of the options given are correct

24. Which of the above is independent of the unit of measurement?

a. i only
b. ii only
c. iii only
d. i and ii only
e. None of the above

25. Graphical scale has more advantages over the other types of scale because?

47. Which of the following equipment could be used to set up right angles in chain surveying?

a. Tilting level
b. Theodolite
c. Cross staff
d. Magnetic Compass
e. Abney Level

48. To measure the slope between two-point A & B, which of the following instruments is used?

a. Cross staff
b. Prism Square
c. Abney level
d. Optical square
e. None of the above

A distance of 159.400 m was measured between two points along a slope of 1 in 5.

Use this information to answer questions 49– 50.

49. The slope angle is?
a. 0.2°
b. 11.31°
c. 0.98°
d. 0.0035°
e. None of the above

50. The horizontal distance between the two points is?
a. 31.880 m.
b. 165.305 m.
c. 156.305 m.
d. 131.880 m
e. None of the above

51. A horizontal distance of 745.000 m is to be established along a line that slopes at a vertical angle of 5° 10'. What slope distance should be measured off?
a. 747.450 m.
b. 747.460 m.
c. 747.400 m.
d. 747.000 m.
e. None of the above

52. The true length of a line is known to be 500 m. The line was again measured with a 20 m tape and found to be 502 m. What is the correct length of the 20 m tape?
a. 20.08m
b. 22.00m
c. 19.92m
d. 25.10m

58. A relatively permanent point of reference whose elevation is known as?
    a. Reduced level
    b. Bench mark
    c. Level surface
    d. Datum point
    e. None of the above

59. The surface tangential to a level surface is said to be a.....
    a. Horizontal surface
    b. Vertical surface
    c. Ground surface
    d. Levelled surface
    e. None of the above

60. Which reading is taken first at a change point?
    a. Backsight
    b. Foresight
    c. Intersight
    d. Reduced level
    e. None of the above

61. 'Change point' in levelling is.....?
    a. an instrument position
    b. a staff position
    c. A back sight position
    d. A fore sight position
    e. none of the above

62. In a levelling operation, 5 back sight readings, 7 intermediate sight readings and 5 foresight readings were taken, how many change point were observed?
    a. 5
    b. 7
    c. 10
    d. 12
    e. None of the above

63. For longer distances, the effect of curvature and refraction will cancel out if ?
    a. The sight distances differ
    b. Back sight distance is greater than that of fore sight
    c. Fore sight distance is greater than that of back sight
    d. Back sight distance is equal to that of fore sight
    e. None of the above

30. One edge of a botanical garden 2½ kilometres long is represented on a map by 5cm. What is the scale of the map?

a. 1:500
b. 1:5000
c. 1:50000
d. 1:500000
e. None of the above

31. Which of the following statements about a map A having a scale of 1: 1000 and another map B having a scale of 1: 5000 is true?
a. Map A is a large-scale map compared to map B
b. Map B is a large-scale map compared to map A
c. Map B is a more detailed map compared to map A
d. Map A and map B are all large-scale maps
e. None of the above

32. A surveyor measures the distance between two points on a map of RF of 1/100 as 60 m, but later finds that he used the wrong RF of 1/50. What is the correct distance between the two points?
a. 30 m
b. 45 m
c. 90 m
d. 120 m
e. None of the above

33. Which of the scale is the smallest one?
a. 4:200000
b. 1 cm = 5000 m
c. 1 cm = 50 m
d. RF = 1/50000
e. None of the above

34. A scale is written as 4cm to 320m, what is its' equivalence among the following?

a. 1 in 8
b. 1 in 80
c. 1 in 8000
d. 1 in 80000
e. None of the above

35. A field distance of 28.45m was represented on a plan by 5.69cm. What is the scale of the plan?

a. 1: 20
b. 1: 200

e. None of the above

53. A survey XY line passes through a pond; the surveyor thus decides to set a perpendicular offset from the line at point A to B of length 53.51m. From B he sets a point on line XY just after the pond and measures it to be 70.15m. What is the length of that part of XY that he cannot measure?

a. 54.36m
b. 45.36m
c. 88.23m
d. 67.42m
e. None of the above

A line AB is measured using a tape of nominal length 50.00 m and was found to be 285.40 m long. When checked against a standard, the tape was found to be 0.05 m longer. Use this information to answer **questions 54 and 55**.

54. What is the correction to the length AB?

a. -0.014 m
b. -0.178 m
c. +0.178 m
d. +0.014 m
e. None of the above

55. What is the correct length of line AB?

a. 285.386 m
b. 285.414 m
c. 285.584 m
d. 285.40 m
e. None of the above

56. Which of the following statements is false?

a. Datum surface is an arbitrary level surface to which the heights of all points are referred
b. A level surface is a curved surface parallel to the mean surface of the earth
c. A level surface is a surface of constant elevation that is perpendicular to a plumb line at every point
d. A level surface is also referred to as a plumb line or a line of gravity
e. None of the above

57. A level surface to which elevation of different points are referred to, is known as?

a. Level surface
b. Datum
c. Horizontal
d. Plane
e. None of the above

64. Which of the following errors can be eliminated by making backsights and foresights equal in length?

a. Staff graduation error
b. Change point error
c. Spirit level error
d. collimation error
e. None of the above

65. Which of the following is/are not (a) source(s) of error in levelling?

I. Non-verticality of staff
II. Unstable grounds
III. Curvature of the earth
IV. Temperature of the staff

a. II only
b. III only
c. II and III only
d. III and IV only
e. None of the above

66. You have undertaken a levelling where you had the chance of setting-up your instrument only three (3) times. You have also reduced all your levels using the Height of Collimation method but you noticed an error of misclosure of +15 cm. Which of the following formulae can help you determine the allowable misclosure and also determine whether or not the level is acceptable?

a. $\pm 5\sqrt{n}$ m and level is acceptable
b. $\pm 5\sqrt{n}$ mm and level is acceptable
c. $\pm 5\sqrt{n}$ m and level is not acceptable
d. $\pm 5\sqrt{n}$ mm and level is not acceptable
e. $\pm 5\sqrt{n}$ cm and level is acceptable

Use the information below to answer questions 67 to 69

From a survey station A, true bearings were taken to two points B and C and the following results obtained.

AB N 00° 00′E
AC S 84° 35' E

If the magnetic bearing of line AB is known to be N 05° 25′ E:

67. What is the angle of declination?

a. 00° 25′ E
b. 05° 25′ E
c. 05° 25′ W
d. 00° 25′ E
e. None of the above

68. Calculate the magnetic bearing of the line AC?

a. N 84° 35′ E

6. To design a planning scheme for a new community in Asuboi, officials require a large-scale map showing all the main physical features (natural and man-made) as well as the relief of the area. Which of the following survey have to be carried out for this project?

    a. Hydrographic Surveying
    b. Cadastral Surveying
    c. Topographic Surveying
    d. Photogrammetric surveying
    e. None of the above

7. The Government of Ghana requires a topographical map of a high order of accuracy over the whole of the Ashanti Region in the shortest possible time. Which survey method is best suited for this project?

    a. Engineering Survey
    b. Remote Sensing
    c. Photogrammetric Survey
    d. Hydrographic Survey
    e. None of the above

8. The Anlo paramount chief desires to know the locations of land parcels and their respective owners in his traditional area. Which kind of map does he need?

    a. Thematic map
    b. Topographic map
    c. Engineering map
    d. Cadastral map
    e. None of the above

9. Which kind of survey would be employed to assess the occurrence of deformation or otherwise of the wall of the Akosombo Dam?

    a. Engineering Survey
    b. Remote Sensing
    c. Photogrammetric Survey
    d. Hydrographic Survey
    e. None of the above

10. When the curvature of the earth is taken into consideration, the surveying is called as?

    a. Plane Surveying
    b. Preliminary Surveying
    c. Hydrographic Surveying
    d. Geodetic Surveying
    e. None of the above

11. Precision is a term that indicates the degree of conformity of:

    a. measured value to its true value.

c. 1: 500
d. 1: 50
e. None of the above

Use the information below to answer questions 36 to 38

A plot of land is in the form of a rectangle in which the length is three times the breadth. When surveyed, it was found to have an area of 24,500m^2. It was then drawn on a plan at a scale of 1:5000.

36. What is the plan area?

a. 0.00098mm^2
b. 980mm^2
c. 9800mm^2
d. 98mm^2
e. None of the above

37. Calculate the dimension of the sides (Length & Breadth respectively) as drawn on the plan?

a. 271.11m & 90.37m
b. 90.37m & 271.11m
c. 18.07mm & 54.21mm
d. 54.21mm & 18.07mm
e. None of the above

38. Calculate the dimension of the sides (Length & Breadth respectively) on the ground?

a. 271.11m & 90.37m
b. 90.37m & 271.11m
c. 18.07mm & 54.21mm
d. 54.21mm & 18.07mm
e. None of the above

39. Suppose you have a map with a scale of 1:50,000. You measure the distance along a property boundary as 2.7 cm. What is the length in the real world?

a. 1.35cm
b. 1.35km
c. 1.35m
d. 13.5m
e. None of the above

40. The method of obtaining horizontal distances directly on the field by dividing the line into segments with the horizontal distance of each segment determined and summed together is known as?

a. Chaining
b. Step chaining
c. Ranging
d. Trilateration
e. None of the above

41. In chain surveying, a river may be considered as an obstacle preventing?

a. Ranging not chaining
b. Chaining not ranging
c. Neither chaining nor ranging
d. Both chaining and ranging
e. None of them

42. In chain survey the area is divided into?

a. Circle
b. Square
c. Triangles
d. Rectangulars
e. None of the above

43. Which of the following is true in step chaining?

a. The length of the step has to be equal between two stations
b. The length of steps which can be adopted is limited by the gradient
c. The tape can be below or above the surveyor's eye level
d. As the gradient increases the length of step also increases
e. None of the above

44. ..... is the lateral measurement taken from an object to the chain line?

a. Taping
b. Offset
c. Detailing
d. All of the alternatives
e. None of the above

45. Offsets are classified according to?

a. size of object and length
b. direction and length
c. direction and shape of object
d. length of chain line and direction
e. None of the above

46. The zero of a tape is held at P and the minimum readings as it crosses the chain line after a swing is at Q. Given that the distance from P to the chain line is less than 7m. What is PQ?

a. PQ is the perpendicular offset of P.
b. PQ is tangential to the chain line.
c. PQ is parallel to the chain line.
d. PQ bisects the chain line.
e. None of the above

BS	IS	FS	Rise	Fall	Initial RL	Adjust-ment	Adjusted RL	Remarks
1.353					47.942	Q73... ...	Q74... ...	BM1 = 47.942m
1.892		Q75...	1.07		Q76...	Q77...	49.0168	1
2.104		0.563	1.329		50.341	0.0096	50.3506	2
	3.553			Q78...	Q79...	Q80...	48.9064	3
	Q81...		0.304		49.196	0.0144	49.2104	4
	3.067		Q82...		Q83...	Q84...	49.3924	5
	2.881		0.186		49.564	Q85...	49.5784	6
	3.115			Q86...	Q87...	0.0144	49.3444	7
	3.113		Q88...		49.332	Q89...	49.3464	8
1.735		Q90	1.27		Q91...	0.0144	50.6164	9
	Q92..			1.493	49.109	0.0192	49.1282	10
	3.225		Q93...		Q94...	0.0192	49.1312	11
3.875		4.221		Q95...	48.116	0.0192	Q96...	12
		Q97	2.693		Q98.....	0.024	50.833	BM2= 50.833m

a	b	c	d	e
0.283	47.942	49.012	0.0048	None of the above
1.449	48.892	0.0144	0.182	None of the above
3.249	0.002	50.602	49.378	None of the above
0	1.843	49.33	0.234	None of the above
3.228	49.112	0.996	0.003	None of the above
48.1352	50.809	1.182	50.807	None of the above

99. What is the misclosure?

a. 0.011m
b. 0.0048m
c. – 0.024m
d. 0.024m
e. None of the above

100. How many change points were used in all?

a. 10
b. 5
c. 3
d. 2
e. 4

E. O Akrofi
C. Y. Asante
S. A. Andam-Akorful
E. V. S. Kwarteng
S. Osah